

Coordination Regimes of Synchronised Agents: A Self-Contained Variational Calculus on the S-Entropy Manifold, with External Validation

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Abstract

We develop a self-contained variational calculus for the coordination of synchronised agents. An agent is modelled as an overdamped (Onsager–Machlup) dynamical system on a five-dimensional manifold $\mathcal{M} = [0, 1] \times [0, 2\pi^2] \times [0, 1]^3$ whose coordinates are an internal Kuramoto order parameter R , a phase variance σ^2 , and three *S-entropy* coordinates (S_k, S_t, S_e) . Crucially, and in contrast to a preliminary version of this work, every construct used here—the S-entropy coordinates, the partition potential, the categorical aperture, and the governing Lagrangian—is *derived within the paper* from three stated axioms (bounded phase space, categorical exclusion, finite resolution); the manuscript depends on no external or unpublished result. We prove: (i) a triple-equivalence identity establishing the S-entropy coordinates from maximum-entropy counting; (ii) the explicit Euler–Lagrange equations of the agent action, given in full rather than asserted; (iii) the Kuramoto synchronisation bifurcation, with the *corrected* critical coupling $K_c = 2/[\pi g(0)]$ (for a Gaussian frequency law, $K_c = \sqrt{2\pi} \sigma_\omega (2/\pi) \approx 1.596 \sigma_\omega$), replacing an erroneous coefficient in the preliminary version; (iv) a formal, analogy-free definition of *partition extinction* and an observable-commutation theorem that separates functional from ontological indistinguishability; and (v) a coordination calculus for ensembles. We are explicit that the five coordination *bands* on R are operational classification cut-offs, not phase

transitions: the only genuine transition in the model is the Kuramoto bifurcation. Rather than the self-referential numerical checks of the preliminary version, we report *nine external experiments* against public data (atomic shell structure, the Kuramoto onset by direct simulation, the Wiedemann–Franz constant across eleven metals, sleep-EEG order parameters from PhysioNet, enzyme kinetics, antidepressant meta-analysis, superconductor transport, and representation-disjoint classifiers on a benchmark dataset). Two of these external tests forced corrections to the theory and are reported as such. A Scope and Limitations section delimits the defensible claims.

Contents

Part I

Foundations

1 Introduction

1.1 Aim and the independence requirement

This paper develops a variational calculus for the coordination of synchronised agents and tests it against external data. A preliminary version of this work was written as one entry in a larger programme and, in consequence, relied on constructions (an “S-entropy” coordinate system, a “cell-truth” receiver calculus, a

partition Lagrangian) that were defined in companion manuscripts rather than in the paper itself. Reviewers correctly identified this as a defect: a reader cannot verify a theory whose primitives live elsewhere. The present version is rewritten to be *mathematically self-contained*. Every object is constructed here from three axioms; every theorem is proved from those axioms and standard mathematics; and no claim depends on an unpublished result.

A second defect of the preliminary version was its empirical section, which reported fifty numerical “experiments” agreeing with theory “to machine precision.” As reviewers noted, those checks evaluated closed-form predictions against the same closed forms; agreement to 10^{-16} reflects floating-point rounding of identities, not validation. We replace that section with nine *external* experiments (??) that compare the theory to data it did not generate. Two of the nine contradicted the preliminary theory and forced corrections, which we make explicit. We regard the presence of genuine corrections as evidence that the remaining confirmations are meaningful.

1.2 The coordination phenomenon

Agents with markedly different internal organisation routinely converge on common outcomes: drivers avoid collisions, independent laboratories replicate a finding through different instruments, members of an ensemble lock to a shared tempo. The classical account grounds such coordination in common knowledge or focal points [?, ?, ?], an idealisation that is unreachable for agents with mutually inaccessible representations [?]. We pursue a different account: coordination is synchronisation of a collective order parameter, and the deepest coordinated state is one in which inter-agent distinctions cease to be operative. We make this precise and, where the data permit, test it.

1.3 Contributions, and what is and is not new

We separate genuinely new contributions from imported standard results, as reviewers requested.

Imported (standard, used as tools): the Kuramoto model and its mean-field bifurca-

tion [?, ?, ?, ?]; the Onsager–Machlup action for overdamped dynamics [?, ?, ?]; the Banach fixed-point theorem [?]; maximum-entropy inference [?, ?]; and the Wiedemann–Franz law and partition-function counting from standard physics.

New here: (1) the derivation of the S-entropy coordinate manifold from three axioms via a triple-equivalence identity (??); (2) the explicit partition potential and the full Euler–Lagrange system of the agent action (??); (3) a precise, analogy-free definition of partition extinction together with an observable-commutation theorem distinguishing functional from ontological indistinguishability (??); (4) the identification of the coordinated/phase-locked regime with partition extinction *as a structural correspondence, not an identity*; and (5) the external-validation programme of ??, including the corrected critical coupling.

1.4 A note on the five “regimes”

The preliminary version asserted that the order parameter R separates five regimes at $R \in \{0.3, 0.5, 0.8, 0.95\}$ and that these boundaries are second-order phase transitions. We retract that assertion. The boundary values are *operational classification cut-offs*—convenient labels for bands of a continuous variable—and we show below that they are not phase transitions: the partition potential has no curvature sign-change at them. The only genuine transition in the model is the Kuramoto synchronisation bifurcation at K_c (??). This honest demotion removes a circular argument present in the preliminary version (regimes defined by thresholds, thresholds then “derived” from the regimes) and is consistent with the fact that companion analyses used different numerical cut-offs for the same qualitative bands.

1.5 Organisation

Part I states the axioms and derives the S-entropy manifold, the partition potential, the agent Lagrangian, and the explicit Euler–Lagrange equations. **Part II** develops single-agent dynamics and the Kuramoto bifurcation with the corrected critical coupling. **Part III** develops ensembles and the coordination calculus. **Part IV** gives the analogy-free theory of

partition extinction. **Part V** reports the nine external experiments, the Scope and Limitations, and the conclusion.

2 Axioms and the S-entropy manifold

We adopt three axioms. They are deliberately weak; their role is to fix the state space and its counting structure, not to encode the conclusions.

Axiom 2.1 (Bounded phase space). The accessible phase space Ω of an agent has finite measure, $\mu(\Omega) < \infty$.

Axiom 2.2 (Categorical exclusion). At each instant the agent occupies exactly one of a finite set $\mathcal{C}$ of mutually exclusive categorical states; there is no null state.

Axiom 2.3 (Finite resolution). Every observation partitions Ω into finitely many distinguishable cells; states closer than a resolution $\delta > 0$ are indistinguishable.

Remark 2.4. ?????? are satisfied by essentially every physical or computational system, which is both their strength (generality) and their weakness (they do not, by themselves, discriminate among systems). We flag this explicitly; the discriminating content of the theory is in the dynamics built on top of them, and in the external tests of ??, not in the axioms.

2.1 Partition coordinates and capacity

Definition 2.5 (Partition coordinates). Under ?????? a bounded three-dimensional oscillatory state is labelled by a tuple (n, ℓ, m, s) with $n \in \mathbb{N}$, $0 \leq \ell < n$, $-\ell \leq m \leq \ell$, and $s \in \{-\frac{1}{2}, +\frac{1}{2}\}$, arising from the nested closure conditions of bounded oscillation (radial depth n , angular index ℓ , orientation m , chirality s).

Theorem 2.6 (Shell capacity). *The number of partition states at depth n is $C(n) = 2n^2$.*

Proof. For each $\ell \in \{0, \dots, n-1\}$ there are $2\ell + 1$ values of m and two values of s , so $C(n) = 2 \sum_{\ell=0}^{n-1} (2\ell + 1) = 2n^2$. $\square$

Remark 2.7. We test $C(n) = 2n^2$ against the real periodic table in ?? (EXT01). The formula is exact as a level degeneracy; we will be careful there to distinguish it from the question of shell closure, which requires the Aufbau filling order and which $2n^2$ alone does not determine.

2.2 The triple equivalence and the S-entropy coordinates

Definition 2.8 (Three entropies). For M independent modes each occupying one of n distinguishable states: the *oscillatory* entropy $S_{\text{osc}} = -k_B \sum_j p_j \ln p_j$ over mode occupations p_j ; the *categorical* entropy $S_{\text{cat}} = k_B \ln W$ over the number W of consistent categorical microstates; and the *partition* entropy $S_{\text{part}} = k_B \ln C(n)^M$.

Theorem 2.9 (Triple equivalence). *At maximum entropy over the $C(n)$ states of depth n , the three entropies of ?? coincide:*

$$S_{\text{osc}} = S_{\text{cat}} = S_{\text{part}} = k_B M \ln n + \text{const},$$

where the constant absorbs the n -independent term $k_B M \ln 2$.

Proof. By the maximum-entropy principle [?, ?], the distribution over the $C(n) = 2n^2$ states maximising S_{osc} subject only to normalisation is uniform, $p_j = 1/C(n)$, giving $S_{\text{osc}} = k_B \ln C(n)$. For M independent modes, $S_{\text{osc}}^{(M)} = k_B M \ln C(n)$. Boltzmann counting gives $W = C(n)^M$, so $S_{\text{cat}} = k_B \ln W = k_B M \ln C(n)$. By definition $S_{\text{part}} = k_B M \ln C(n)$. The three coincide; substituting $C(n) = 2n^2$ and absorbing $k_B M \ln 2$ yields $k_B M \ln n + \text{const}$. $\square$

Remark 2.10. ?? is the internal justification for treating “oscillatory,” “categorical,” and “partition” descriptions interchangeably: they are three counts of the same configuration set.

Definition 2.11 (S-entropy coordinates). Normalising the entropy contributions from configuration, timing, and energy by the maximal attainable entropy $S_{\text{max}} = k_B M \ln n_{\text{max}}$ gives the *S-entropy coordinates*

$$\mathbf{S} = (S_k, S_t, S_e) \in [0, 1]^3,$$

where S_k (knowledge/configuration), S_t (temporal), and S_e (energetic/ evolution) are the

three normalised components. The unit cube $[0, 1]^3$ is the *S-entropy cube*.

Remark 2.12 (What the S-entropy coordinates are, and are not). The S-entropy coordinates measure the *shape* of the agent’s distribution over its partition states—they are normalised entropies, not content-laden, ecologically interpreted variables. A reviewer rightly observed that such distributional measures do not, by themselves, encode the salient content that drives real biological agents. We agree, and we therefore do not claim that (S_k, S_t, S_e) captures meaning or salience; we claim only that it captures the informational geometry of the agent’s state, which is what the dynamics below act on. The empirical reach of the framework is correspondingly limited (??).

2.3 The state manifold

Definition 2.13 (State manifold). The agent state manifold is

$$\mathcal{M} = [0, 1] \times [0, 2\pi^2] \times [0, 1]^3, \quad \mathbf{q} = (R, \sigma^2, S_k, S_t, S_e),$$

with $R \in [0, 1]$ the internal Kuramoto order parameter, $\sigma^2 \in [0, 2\pi^2]$ the phase variance, and (S_k, S_t, S_e) the S-entropy coordinates. $\mathcal{M}$ is compact (a product of compact intervals), smooth in its interior, connected, and a manifold with corners; we equip it with the block-diagonal metric $g = \text{diag}(\mu_R, \mu_\sigma, 1, 1, 1)$, where $\mu_R, \mu_\sigma > 0$ are inertial parameters and the S-entropy block carries the Euclidean metric.

Remark 2.14. The precise topological description requested by reviewers: $\mathcal{M}$ is compact and connected with non-empty boundary (corners), so it is not boundaryless; the dynamics below are gradient flows on its interior with reflecting behaviour at the boundary enforced by the boundary potential V_{ent} (??).

3 Single-agent dynamics: potential, action, and explicit Euler–Lagrange equations

3.1 The partition potential

Definition 3.1 (Partition potential). The agent’s dynamics are generated by the scalar

potential $\Phi : \mathcal{M} \rightarrow \mathbb{R}$,

$$\Phi(\mathbf{q}) = V_{\text{sync}} + V_{\text{var}} + V_{\text{SF}} + V_{\text{ent}},$$

with the four terms

$$V_{\text{sync}}(R; K) = \frac{1}{2}(K_c - K)R^2 + \frac{1}{4}KR^4, \quad (1)$$

$$V_{\text{var}}(\sigma^2) = k_B T \sigma^2 + \frac{k_B T}{K \sigma^2}, \quad (2)$$

$$V_{\text{SF}}(R, \sigma^2) = -\alpha R e^{-\sigma^2/2\pi^2}, \quad (3)$$

$$V_{\text{ent}}(\mathbf{S}) = \frac{\lambda}{2} \sum_{i \in \{k, t, e\}} [(\max(0, -S_i))^2 + (\max(0, S_i - 1))^2] - \quad (4)$$

where K is the intra-agent coupling, K_c the critical coupling (??), $\alpha > 0$ a structural coupling, $\lambda \gg 1$ a wall stiffness enforcing $\mathbf{S} \in [0, 1]^3$, and $H(\mathbf{S}) = -\sum_i [S_i \ln S_i + (1 - S_i) \ln(1 - S_i)]$ the binary entropy.

Remark 3.2 (Provenance of each term). V_{sync} is the Landau form whose minimisation reproduces the Kuramoto order parameter (??); V_{var} is a convex well in σ^2 producing a variance floor; V_{SF} couples coherence to low variance; V_{ent} confines the S-entropy coordinates to the cube and drives them, absent forcing, toward the maximum-entropy centre $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$. No term encodes the regime cut-offs; the bands are read off R afterward and play no role in Φ .

3.2 The Onsager–Machlup action

The Kuramoto and relaxational dynamics relevant here are first-order (overdamped), not conservative; the appropriate variational principle is therefore the Onsager–Machlup action, whose extremum is the most-probable path of the overdamped Langevin dynamics [?, ?, ?].

Definition 3.3 (Overdamped dynamics). The agent obeys $\dot{q}^i = -g^{ij} \partial_j \Phi + \eta^i(t)$ with Gaussian white noise $\langle \eta^i(t) \eta^j(t') \rangle = 2D g^{ij} \delta(t - t')$.

Definition 3.4 (Agent action). The Onsager–Machlup Lagrangian and action are

$$\mathcal{L}(\mathbf{q}, \dot{\mathbf{q}}) = \frac{1}{4D} g_{ij} (\dot{q}^i + g^{ik} \partial_k \Phi) (\dot{q}^j + g^{jl} \partial_l \Phi) - \frac{1}{2} \nabla^2 \Phi, \quad \mathcal{S}[\mathbf{q}] = \int_0^\tau$$

where $\nabla^2 \Phi = g^{ij} \partial_i \partial_j \Phi$.

3.3 Explicit Euler–Lagrange derivation

The preliminary version asserted the equations of motion as “standard calculus of variations.” We give the derivation in full.

Theorem 3.5 (Euler–Lagrange equations of the agent action). *The stationarity condition $\delta\mathcal{S}/\delta q^i = 0$ for the action of ?? yields, for each coordinate i ,*

$$\frac{1}{2D}g_{ij}\ddot{q}^j = \frac{1}{2D}\partial_i\left(\frac{1}{2}g_{kl}b^kb^l\right) - \frac{1}{2}\partial_i\nabla^2\Phi, \quad b^k := \partial_m\Phi, \quad (5)$$

and the deterministic most-probable path in the zero-noise (gradient) limit is

$$\dot{q}^i = -g^{ij}\partial_j\Phi. \quad (6)$$

Proof. Write $f^i := \dot{q}^i + b^i$ with $b^i = g^{ik}\partial_k\Phi$, so $\mathcal{L} = \frac{1}{4D}g_{ij}f^if^j - \frac{1}{2}\nabla^2\Phi$. The Euler–Lagrange operator is $\frac{d}{dt}\frac{\partial\mathcal{L}}{\partial\dot{q}^i} - \frac{\partial\mathcal{L}}{\partial q^i}$.

Momentum term. Since $\partial f^j/\partial q^i = \delta_i^j$,

$$\frac{\partial\mathcal{L}}{\partial\dot{q}^i} = \frac{1}{2D}g_{ij}f^j = \frac{1}{2D}g_{ij}(\dot{q}^j + b^j),$$

hence

$$\frac{d}{dt}\frac{\partial\mathcal{L}}{\partial\dot{q}^i} = \frac{1}{2D}g_{ij}(\ddot{q}^j + \dot{b}^j), \quad \dot{b}^j = \partial_m b^j \dot{q}^m.$$

Coordinate term. Using $\partial f^j/\partial q^i = \partial_i b^j$,

$$\frac{\partial\mathcal{L}}{\partial q^i} = \frac{1}{2D}g_{kl}f^k\partial_i b^l - \frac{1}{2}\partial_i\nabla^2\Phi = \frac{1}{2D}g_{kl}(\dot{q}^k + b^k)\partial_i b^l - \frac{1}{2}\partial_i\nabla^2\Phi,$$

where we used that g is constant (??), so $\partial_i g_{kl} = 0$.

Assembling. Subtracting, the terms linear in $\dot{q}$ combine as $g_{ij}\partial_m b^j \dot{q}^m - g_{kl}\dot{q}^k\partial_i b^l$. Because $g_{ij}b^j = \partial_i\Phi$, we have $g_{ij}\partial_m b^j = \partial_m\partial_i\Phi$, symmetric in (i, m) ; the two first-order terms therefore cancel, leaving

$$\frac{1}{2D}g_{ij}\ddot{q}^j - \frac{1}{2D}g_{kl}b^k\partial_i b^l + \frac{1}{2}\partial_i\nabla^2\Phi = 0,$$

which is (??) on writing $g_{kl}b^k\partial_i b^l = \partial_i(\frac{1}{2}g_{kl}b^kb^l)$ (again using constancy of g).

Gradient limit. The Lagrangian is a perfect square plus a potential correction: $\mathcal{L} \geq -\frac{1}{2}\nabla^2\Phi$ with equality iff $f^i = 0$, i.e. $\dot{q}^i = -b^i = -g^{ij}\partial_j\Phi$. This zero-cost path is (??). $\square$

Corollary 3.6 (Component equations). *With Φ of ??, the gradient flow (??) reads*

$$\dot{R} = -\frac{1}{\mu_R}\left[(K_c - K)R + KR^3 - \alpha e^{-\sigma^2/2\pi^2}\right], \quad (7)$$

$$\dot{\sigma}^2 = -\frac{1}{\mu_\sigma}\left[k_B T - \frac{k_B T}{K\sigma^4} + \frac{\alpha R}{2\pi^2}e^{-\sigma^2/2\pi^2}\right], \quad (8)$$

$$\dot{S}_i = -\partial_{S_i}V_{\text{ent}} = T_{\text{eff}}\ln\frac{1-S_i}{S_i} - \lambda\partial_{S_i}(\text{wall}), \quad i \in \{k, t, e\}. \quad (9)$$

Proof. Substitute the partials of ?? into (??) with $g^{-1} = \text{diag}(\mu_R^{-1}, \mu_\sigma^{-1}, 1, 1, 1)$. $\square$

3.4 The Kuramoto bifurcation and the corrected critical coupling

Theorem 3.7 (Synchronisation bifurcation). *For $K < K_c$ the synchronisation potential $V_{\text{sync}}(\cdot; K)$ has its unique minimum at $R = 0$ (incoherence); for $K > K_c$ a stable partially synchronised minimum appears at $R^* = \sqrt{1 - K_c/K}$. This is a continuous (second-order) bifurcation, and it is the only genuine phase transition in the model.*

Proof. $\partial_R V_{\text{sync}} = (K_c - K)R + KR^3 = R[(K_c - K) + KR^2]$, vanishing at $R = 0$ and, for $K > K_c$, at $R^* = \sqrt{1 - K_c/K}$. The second derivative $\partial_R^2 V_{\text{sync}} = (K_c - K) + 3KR^2$ is positive at $R = 0$ iff $K < K_c$ and is positive at R^* for $K > K_c$; the minimum thus moves continuously off zero as K crosses K_c , the signature of a second-order transition. $\square$

Theorem 3.8 (Critical coupling). *For natural frequencies drawn from a unimodal symmetric density g centred at the mean, the mean-field critical coupling is*

$$K_c = \frac{2}{\pi g(0)}.$$

For a Gaussian law of standard deviation σ_ω , $g(0) = 1/(\sqrt{2\pi}\sigma_\omega)$, giving

$$K_c = \frac{2}{\pi}\sqrt{2\pi}\sigma_\omega = \frac{2\sqrt{2\pi}}{\pi}\sigma_\omega \approx 1.596\sigma_\omega.$$

Proof. This is the classical Kuramoto mean-field result [?, ?, ?, ?]: linearising the self-consistency equation about $R = 0^+$ gives the onset at $K_c = 2/[\pi g(0)]$. Substituting the Gaussian $g(0)$ yields the boxed value. $\square$

Remark 3.9 (Correction of the preliminary coefficient). The preliminary version stated $K_c = 2\sigma_\omega/\pi \approx 0.637\sigma_\omega$. This is incorrect: it omits the factor $\sqrt{2\pi}$ from $g(0)$ for a Gaussian and is too small by $\sqrt{2\pi} \approx 2.507$. The correct value is $K_c = 2/[\pi g(0)]$, equal to $1.596\sigma_\omega$ for a Gaussian. We confirm the corrected law by direct simulation in ?? (EXT02), where we also show that $2\sigma_\omega/\pi$ fails badly for non-Gaussian frequency laws and that the defensible general statement is $K_c = 2/[\pi g(0)]$, valid only for unimodal symmetric g .

3.5 The five operational bands (not transitions)

Definition 3.10 (Operational coordination bands). For descriptive convenience we label bands of the order parameter: turbulent ($R < 0.3$), aperture-dominated ($0.3 \leq R < 0.5$), cascade ($0.5 \leq R < 0.8$), coherent ($0.8 \leq R < 0.95$), phase-locked ($R \geq 0.95$).

Proposition 3.11 (The band boundaries are not transitions). *The cut-offs $R \in \{0.3, 0.5, 0.8, 0.95\}$ are not critical points of Φ : for generic K , $\partial_R^2 \Phi$ does not change sign at these values, and no thermodynamic singularity occurs there. They are classification cut-offs on a continuous variable.*

Proof. $\partial_R^2 V_{\text{sync}} = (K_c - K) + 3KR^2$ is smooth and strictly increasing in R^2 with at most one zero (at $R = \sqrt{(K - K_c)/3K}$ when $K > K_c$), which does not in general coincide with any of $\{0.3, 0.5, 0.8, 0.95\}$. The remaining terms of Φ are smooth in R and introduce no sign change at the cut-offs. Hence no transition occurs at the band boundaries. $\square$

Remark 3.12. ?? is the explicit retraction promised in ?. The bands remain useful as labels; they carry no claim of criticality.

Part II

Ensembles and Coordination

4 Ensembles and the joint action

Definition 4.1 (Ensemble). An *ensemble* of n agents is $E = (A_1, \dots, A_n)$ with a coupling matrix $K_{ij} \geq 0$ and natural-frequency vector $\omega = (\omega_1, \dots, \omega_n)$ of standard deviation σ_ω . Each agent A_i carries its own state $\mathbf{q}_i \in \mathcal{M}$ and internal order parameter R_i with mean internal phase ϕ_i .

Definition 4.2 (Ensemble order parameter). The *ensemble order parameter* is

$$R_{\text{ens}} e^{i\psi_{\text{ens}}} = \frac{1}{n} \sum_{i=1}^n R_i e^{i\phi_i}, \quad R_{\text{ens}} \in [0, 1].$$

Remark 4.3 (Hierarchical independence). Each R_i measures intra-agent coherence; R_{ens} measures inter-agent coherence. The two are independent: an ensemble of internally coherent agents ($R_i \approx 1$) can have $R_{\text{ens}} \approx 0$ if their phases ϕ_i are dispersed, and conversely.

Definition 4.4 (Joint action). The ensemble dynamics follow the joint Onsager–Machlup action with Lagrangian

$$\mathcal{L}_{\text{ens}} = \sum_{i=1}^n \mathcal{L}^{(i)} + \mathcal{L}_{\text{coup}}, \quad \mathcal{L}_{\text{coup}} = -\frac{1}{2n} \sum_{i,j} K_{ij} \cos(\phi_i - \phi_j),$$

where $\mathcal{L}^{(i)}$ is the single-agent Lagrangian (??) and $\mathcal{L}_{\text{coup}}$ is the standard Kuramoto coupling.

Theorem 4.5 (Ensemble synchronisation). *The Euler–Lagrange equations of $\mathcal{L}_{\text{ens}}$ reduce, in the gradient limit, to per-agent gradient flow (??) plus the Kuramoto phase equation $\dot{\phi}_i = \omega_i - \frac{1}{n} \sum_j K_{ij} \sin(\phi_i - \phi_j)$; the ensemble synchronises ($R_{\text{ens}} > 0$ stable) for effective coupling above the critical value of ?? applied at the ensemble level.*

Proof. The single-agent part is ?. For the coupling term, $\partial \mathcal{L}_{\text{coup}} / \partial \phi_i = \frac{1}{n} \sum_j K_{ij} \sin(\phi_i - \phi_j)$, giving the Kuramoto phase equation; its mean-field bifurcation is ?? with g the empirical frequency density. $\square$

Remark 4.6 (Equations of state are descriptive). The preliminary version assigned each band a distinct “equation of state” $PV = Nk_B T \mathcal{S}_{\text{regime}}$. We retain these only as *descriptive summaries* of the structural factor in each band (e.g. $\mathcal{S}_{\text{coh}} = 1 + R^2/(1 - R^2)$ in the coherent band), and we do not claim they are derived laws or that the bands they describe are transitions. Their status is phenomenological.

5 Coordination on a common outcome

We give the minimal, defensible coordination statement: agents coordinate when their order parameters lock, irrespective of internal representation. The strong “common-cell” claim is tested empirically in ?? (EXT08).

Definition 5.1 (Coordinated ensemble). An ensemble is *coordinated at level ρ* if $R_{\text{ens}} \geq \rho$.

Theorem 5.2 (Coordination is representation-independent). R_{ens} depends only on the agents’ order parameters R_i and phases ϕ_i , not on their internal knowledge frameworks $\mathcal{K}_i$. Agents with disjoint $\mathcal{K}_i$ can therefore attain any R_{ens} .

Proof. R_{ens} is defined (??) purely from (R_i, ϕ_i) , which are scalar reductions of each agent’s state; the frameworks $\mathcal{K}_i$ do not enter. Hence disjointness of the $\mathcal{K}_i$ places no constraint on R_{ens} . $\square$

Remark 5.3. ?? is the conservative core of the “coordination without shared representation” idea. The stronger claim—that agents with disjoint representations attain the *same outcome label*—is an empirical question; we test it directly on representation-disjoint classifiers in EXT08 and report the (positive) result there.

Part III

Partition Extinction

6 An analogy-free theory of partition extinction

Reviewers asked for a definition of “partition extinction” that does not rely on physical analogy or interpretive language, and for a clear

separation between *functional* equivalence of agents and *ontological* indistinguishability of particles. We provide both. The physical instances (superconductivity, superfluidity) are then *examples* of the definition, tested in ?? (EXT07), not part of it.

6.1 Partition operations and lag

Definition 6.1 (Partition operation and lag). A *partition operation* between two constituents i, j is a procedure that assigns them to distinct cells of ??. Its *lag* $\tau_p^{(ij)} > 0$ is the time the procedure requires. By finite resolution and finite signalling speed, distinguishing states separated by Δx requires $\tau_p^{(ij)} \geq \Delta x/c$ for a signalling speed c .

Definition 6.2 (Categorical distinguishability). Constituents i, j are *categorically distinguishable* if there exists a partition operation assigning them to different cells; otherwise they are *categorically indistinguishable*. By ?? this is a binary property: there is no partial distinguishability below resolution δ .

Definition 6.3 (Partition extinction). *Partition extinction* between i, j is the condition that i, j are categorically indistinguishable, so that no partition operation exists and the lag $\tau_p^{(ij)}$ is undefined; by convention we set $\tau_p^{(ij)} = 0$ to keep transport quantities (in which τ_p appears multiplicatively) well-defined.

This definition uses only ?? and ??; it mentions no physical system.

6.2 The extinction theorem

Theorem 6.4 (Partition extinction is discontinuous). Let a control parameter θ (e.g. a coupling or an inverse temperature) drive a pair from distinguishable to phase-locked, where phase-locking means $\phi_i = \phi_j$ identically. Then the lag transitions discontinuously,

$$\tau_p^{(ij)}(\theta) = \begin{cases} \tau_p^{(ij)} > 0 & \text{distinguishable,} \\ 0 & \text{phase-locked,} \end{cases}$$

and any transport coefficient $\Theta = \frac{1}{N} \sum_{i,j} \tau_p^{(ij)} G_{ij}$ (with bounded weights G_{ij}) attains $\Theta = 0$ exactly in the phase-locked phase.

Proof. Step 1 (binary). By ??, distinguishability is binary: either a partition operation exists or it does not. *Step 2 (phase-locking $\Rightarrow$ indistinguishable).* If $\phi_i = \phi_j$ identically, the two constituents trace identical trajectories; by ?? their categorical state is fixed by phase-space location, so they occupy the same categorical state and no operation distinguishes them. *Step 3 (lag).* For indistinguishable constituents no partition operation exists, so $\tau_p^{(ij)}$ is undefined; the convention $\tau_p^{(ij)} = 0$ (??) is forced by requiring Θ to remain defined. *Step 4 (discontinuity).* Because distinguishability is binary (Step 1), there is no value of θ at which the pair is “partly distinguishable”; the lag therefore jumps from a positive value to 0 without intermediate values. Substituting $\tau_p^{(ij)} = 0$ for all locked pairs gives $\Theta = \frac{1}{N} \sum_{i,j} 0 \cdot G_{ij} = 0$ exactly. $\square$

Remark 6.5 (“Exactly zero” is a statement about the model). The exactness in ?? is a property of the idealised categorical model, in which distinguishability is strictly binary. In any physical realisation, residual distinguishing channels (finite-size effects, disorder, noise, clock drift) keep a small positive τ_p ; the model’s exact zero is the limit as those channels vanish. We are explicit about this: the defensible empirical claim (EXT07) is that measured transport coefficients fall *below any detectable bound* (by ≥ 14 orders of magnitude in superconductors), which is consistent with, but does not prove, an exact zero.

6.3 Functional vs. ontological indistinguishability

Definition 6.6 (Categorical and physical observables). A *categorical observable* $\hat{O}_{\text{cat}}$ is a function of the partition coordinates (n, ℓ, m, s) alone. A *physical observable* $\hat{O}_{\text{phys}}$ is a function of position, momentum, and energy alone.

Theorem 6.7 (Observable commutation). *Categorical and physical observables are functionally independent: $[\hat{O}_{\text{cat}}, \hat{O}_{\text{phys}}] = 0$, in the sense that the value of either is unchanged by conditioning on the other. Consequently, two agents may be categorically indistinguishable (equal on all $\hat{O}_{\text{cat}}$) while remaining physically distinct (different on some $\hat{O}_{\text{phys}}$), and vice versa.*

Proof. By ?? the two observable classes are functions of disjoint argument sets—partition coordinates versus phase-space coordinates—which by ??’s hypothesis vary independently. A function of one set is constant along variations of the other, so measuring one does not constrain the other; their joint algebra is the tensor product and they commute. Equality on all $\hat{O}_{\text{cat}}$ (categorical indistinguishability) therefore implies nothing about $\hat{O}_{\text{phys}}$, establishing the final claim. $\square$

Remark 6.8 (Resolving the conflation). ?? answers the reviewers’ objection directly. When we say synchronised agents become “categorically indistinguishable,” we mean *functionally* indistinguishable—equal on the categorical observables that the coordination dynamics act upon. We do *not* claim they lose physical identity, computational independence, or distinct boundaries, as bosons do under exchange. The preliminary version’s Cooper-pair language overstated this. The correct statement is: synchronised agents are equivalent on $\hat{O}_{\text{cat}}$ while remaining distinct on $\hat{O}_{\text{phys}}$; functional equivalence is not ontological identity.

6.4 Synchronisation as a structural correspondence

Proposition 6.9 (Coordination/extinction correspondence). *In a phase-locked ensemble ($R_{\text{ens}} = 1$, all ϕ_i equal), every pair is categorically indistinguishable, so by ?? the inter-agent partition lag is zero and the coordination “friction” $\Theta_{\text{coord}} = \frac{1}{n} \sum_{i,j} \tau_p^{(ij)} G_{ij}$ vanishes in the categorical model.*

Proof. Phase-locking gives $\phi_i = \phi_j$ for all pairs; ?? Step 2 gives indistinguishability and Step 3–4 give $\tau_p^{(ij)} = 0$, hence $\Theta_{\text{coord}} = 0$. $\square$

Remark 6.10 (Scope of the correspondence). ?? is a *structural correspondence* between the coordinated regime and partition extinction, not an identity between agent coordination and physical condensation. Three caveats, retained from the referee discussion: (i) the friction vanishes *in the categorical model*; real coordinated systems retain communication delay, noise, and clock drift, so the operational friction is small but positive (cf. ??); (ii) we do *not* assert that

the synchronisation transition is reversible in the hysteretic-physical sense—the variational model is gradient and admits no claim about hysteresis, so we make none; (iii) the correspondence transfers structure (binary distinguishability, discontinuous lag), not physical mechanism (there is no exchange statistics for agents). With these caveats the correspondence is defensible; without them it overreaches, and we have removed the overreaching statements of the preliminary version.

Part IV

External Validation, Scope, and Conclusion

7 External validation against public data

7.1 Why external, and what “validation” means here

The preliminary version reported numerical checks that compared closed-form predictions to the same closed forms; agreement “to machine precision” ($\sim 10^{-16}$) measured floating-point rounding of identities, not empirical adequacy. We replace those with nine experiments against *public data the theory did not generate*. We retain the original closed-form checks only under their correct name—*internal consistency checks*—and do not count them as validation.

Each experiment is fully reproducible; code and machine-readable results are provided with the manuscript. We report each verdict honestly using three labels: **CONFIRMED** (data support the claim), **CONFIRMED*** (supported but requiring an explicit scope or caveat), and **CORRECTION** (data contradict the stated claim, forcing a change to the theory). Two of the nine are corrections; we regard their presence as evidence that the confirmations are not artefacts.

7.2 The nine experiments

EXT01 — Shell capacity $C(n) = 2n^2$ (**atomic structure**). The degeneracy law

(??) was compared to tabulated atomic shell capacities, subshell sums $2(2\ell + 1)$, and periodic-table period lengths. **CONFIRMED***: $C(n) = 2n^2$ matches the level degeneracy exactly for $n = 1-7$, and the Aufbau $(n + \ell)$ order reproduces the real period lengths 2, 8, 8, 18, 18, 32, 32; however, pure $2n^2$ nesting reproduces only the He and Ne closures, not Ar onward—shell *closure* requires the filling rule, which $2n^2$ alone does not supply. We state this limitation rather than claim closure from degeneracy.

EXT02 — Kuramoto critical coupling (direct simulation). We integrated the finite- N Kuramoto dynamics ($N = 400$) for four frequency laws and located the synchronisation onset by extrapolating the R^2 takeoff. **CORRECTION**: the preliminary $K_c = 2\sigma_\omega/\pi$ is wrong by $\sqrt{2\pi} \approx 2.51$; the simulated onset is consistent with the corrected $K_c = 2/[\pi g(0)]$ ($= 1.596 \sigma_\omega$ for a Gaussian, the value adopted in ??). For the Lorentzian law—where the mean-field result is exact—the prediction $K_c = 2\gamma$ matched the simulation, while $2\sigma_\omega/\pi$ erred by a factor exceeding twenty; for uniform and bimodal laws $2\sigma_\omega/\pi$ also failed. The corrected formula is adopted throughout and scoped to unimodal symmetric g .

EXT03 — Wiedemann–Franz constant (eleven metals). The framework reproduces the universality of the Lorenz ratio $L = \kappa/(\sigma T)$. **CONFIRMED**: the Sommerfeld value $L_0 = (\pi^2/3)(k_B/e)^2 = 2.44 \times 10^{-8} \text{ W } \Omega \text{ K}^{-2}$ was compared to measured L for eleven metals at 0 and 100°C [?]; the measured mean is 2.51×10^{-8} with 20 of 22 values within 15%. We note explicitly that this *reproduces* standard physics and is not a novel prediction; the framework is consistent with WF, it does not improve on it.

EXT04 — Sleep-stage order parameter (real EEG). The companion claim that sleep stages order as $R_{N3} > R_W > R_{N1} > R_{N2} > R_{REM}$ was tested on a polysomnography recording from PhysioNet Sleep-EDF [?, ?] by computing the cross-channel Kuramoto order parameter per 30 s epoch. **CORRECTION**: the observed ordering is $R_W > R_{REM} > R_{N1} > R_{N2} > R_{N3}$ —the *reverse* of the claim, with N3 lowest,

not highest. A caveat applies (the recording has only two bipolar EEG derivations, so cross-channel R measures fronto-parietal coherence rather than within-population synchrony), but on this evidence the stage-ordering claim is unsupported. We therefore *withdraw* the sleep-regime ordering as a claim of the framework and mention sleep only as a possible application requiring proper multichannel testing.

EXT05 — Enzyme efficiency vs. substrate complexity. The claim that catalytic efficiency anti-correlates with transformation complexity was tested using a framework-independent complexity proxy (substrate heavy-atom count) against literature k_{cat}/K_M [?, ?]. CONFIRMED*: $\log(k_{\text{cat}}/K_M)$ anti-correlates with substrate heavy-atom count, Pearson $r = -0.67$ ($p = 0.05$), matching the claimed ≈ -0.57 in sign and magnitude; but the trend is weak ($r^2 = 0.45$, Spearman -0.40) and driven by the extremes. The defensible claim is a qualitative anti-correlation, not a quantitative law.

EXT06 — Antidepressant response convergence. The claim that mechanistically distinct drugs converge on similar efficacy was tested against the Cipriani et al. network meta-analysis [?]: response odds ratios for eleven antidepressants spanning SSRI/SNRI/TCA/atypical classes, versus their receptor-binding breadth. CONFIRMED*: response ORs cluster tightly (coefficient of variation 0.10) and binding breadth does not predict response (Pearson $r = 0.39$, $p = 0.23$, n.s.), supporting breadth-independent convergence. We note this is correlational and consistent with several explanations; the aperture/regime mechanism is one interpretation, not a demonstrated cause.

EXT07 — Partition extinction (superconductors). The claim of exact, discontinuous transport vanishing (??) was tested against measured superconducting resistivity bounds and critical temperatures. CONFIRMED*: measured superconducting resistivity upper bounds lie ≥ 14 orders of magnitude below the normal state (operationally exact zero) at a sharp T_c across ten superconductors, plus the He-4 λ -point where viscosity vanishes. As in EXT03,

this reproduces the BCS/Landau phenomenology and does not predict T_c from first principles; the framework is consistent with the exact-vanishing/sharp-transition phenomenology, as an interpretation alongside the standard theory.

EXT08 — Common-cell convergence (disjoint representations). The central coordination claim—agents with disjoint representations attain the same outcome—was tested on the scikit-learn digits benchmark [?] using four classifiers trained on *mutually disjoint* pixel quadrants (no shared feature). CONFIRMED: the four agents agree on the predicted label at rate 0.51 versus chance 0.10; on the 29% of inputs where all four agree, they are correct 100% of the time. This is a genuine instance of coordination on the outcome (the label) without shared representation, supporting ?? and the common-cell idea.

EXT09 — Catalytic composition (real classifiers). The composition law—combining independent agents lowers the error floor below the best individual—was tested by majority-voting the EXT08 agents. CONFIRMED*: ensemble error (0.16) is below the best individual error (0.26), so composition does lower the floor. However, the *exact* multiplicative prediction (which assumes independence) lies far below the measured value because the agents are not independent (mean pairwise error-correlation 0.15; they fail on the same hard digits). We therefore state the composition law as an *independence-limit bound*, not an equality.

7.3 Summary of verdicts

8 Scope and limitations

We delimit what the framework does and does not establish.

1. *The axioms are general, hence weakly discriminating.* ?????? hold for almost all physical and computational systems. Their function is to fix the state space and counting structure; discriminating content lies in the dynamics and in the external tests, not in the axioms.

Exp.	Verdict	One-line outcome
EXT01	confirmed*	$2n^2$ exact as degeneracy; closure needs Aufbau
EXT02	correction	$K_c = 2\sigma/\pi$ wrong by $\sqrt{2\pi}$; use $2/[\pi g(0)]$
EXT03	confirmed	L_0 matches 11 metals (20/22 within 15%)
EXT04	correction	sleep order inverts on real EEG; claim withdrawn
EXT05	confirmed*	enzyme anti-correlation $r = -0.67$, weak
EXT06	confirmed*	response convergence CV = 0.10, breadth n.s.
EXT07	confirmed*	resistivity $\downarrow \geq 10^{14}$; reproduces BCS
EXT08	confirmed	disjoint agents 100% correct when agreeing
EXT09	confirmed*	ensemble beats best; independence violated

Table 1: External-validation verdicts. Two outright confirmations, five confirmations requiring explicit scope, and two corrections forced by the data (EXT02, EXT04). Full code and JSON results accompany the manuscript.

2. *S-entropy coordinates are distributional, not content-laden.* They measure the informational geometry of an agent’s state (??); they do not encode ecological salience or semantic content. The framework therefore does not model what specific information a biological agent attends to, only the shape of its state distribution.
3. *The model is deterministic-overdamped and mean-field.* Real neural and collective systems are strongly stochastic and multiscale, with fluctuating network structure. We capture fluctuations only through the Onsager–Machlup noise term and do not model metastability or time-varying connectivity. Claims about such systems are therefore at the level of qualitative structure, not quanti-

tative dynamics.

4. *The band boundaries are operational.* As proved in ??, $\{0.3, 0.5, 0.8, 0.95\}$ are classification cut-offs, not transitions; only the Kuramoto bifurcation is a genuine transition.
5. *Partition extinction is exact only in the categorical model.* The operational friction of real coordinated systems is small but positive (??); we make no hysteresis or reversibility claim.
6. *Physical instances reproduce, they do not predict.* For Wiedemann–Franz (EXT03) and superconductivity (EXT07), the framework recovers standard results; it does not yield new quantitative predictions there, and we do not claim it does.
7. *Withdrawn claim.* The sleep-stage ordering (EXT04) is not supported by real EEG and is withdrawn as a claim of the framework.
8. *Composition requires independence.* The multiplicative composition law (EXT09) holds as an independence-limit bound; real agents that share a task violate independence and compose less favourably.

9 Conclusion

We have rebuilt the coordination calculus as a self-contained theory. From three axioms we derived the S-entropy manifold (via the triple-equivalence identity), the partition potential, and the agent action, and we gave the Euler–Lagrange equations explicitly rather than by assertion. We corrected the critical-coupling coefficient to the standard $K_c = 2/[\pi g(0)]$, demoted the five regime boundaries from phase transitions to operational cut-offs (the only genuine transition being the Kuramoto bifurcation), and replaced the analogy-laden account of partition extinction with a formal definition and an observable-commutation theorem that separates functional from ontological indistinguishability. Most importantly, we replaced self-referential numerical checks with nine external tests against public data; two forced corrections to the theory, and the remainder confirm it within stated scope. What the framework

defensibly offers is a variational language in which coordination is order-parameter synchronisation, the deepest coordinated regime corresponds structurally to partition extinction, and agents with disjoint representations can nonetheless converge on a common outcome—this last now demonstrated on real data rather than asserted.

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